

PV102 Day 4

Module Behavior String Design and Shade

String design follows limits, temperature, and shade reality.

CTS Day 20 | Thursday, August 6, 2026 | 10:00 AM-3:00 PM

Today’s job

Professional solar work starts before parts move.

• Students connect PV101 series/parallel measurement habits to PV102 design judgment: string voltage, current, controller limits, temperature, and shade-sensitive layout decisions.

- Carry forward the Day 1 habit: match architecture to real need.
- Today adds one design layer and one clean handoff.

Do not guess when an unknown can be recorded.



String design follows limits, temperature, and shade reality.

Learning targets

By 3:00 PM students should be able to:

- Predict how series and parallel wiring affect PV string voltage and current.
- Explain why controller input limits control string design.
- Recognize shade as both a production and mismatch problem.
- Use a safe paper/string model to compare design options before hardware work.



Evidence beats confidence. A tragic lesson for civilization, apparently.

String design follows limits, temperature, and shade reality.

Morning readiness check

Quiz first, then review as evidence.

- Work independently for 12-15 minutes.
- Use wrong answers to locate teaching gaps.
- Ask: what fact would prove this answer?
- Keep review short and specific.



The quiz measures instruction as much as students.

Core rule

- String design follows limits, temperature, and shade reality.
- Say it once at the start.
- Bring every debate back to it.
- Use it to reject weak handoffs.



One rule, repeated until reality behaves.

String design follows limits, temperature, and shade reality.

What students already know

PV101 and RCS are not separate islands.

- PV modules produce DC power.
- Series and parallel change voltage/current behavior.
- Continuity requires de-energized verification.
- Troubleshooting starts from evidence.
- Handoffs protect the next person.



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New judgment today

The skill moves from “can I build it?” to “should this be the design?”

- Define the job.
- Gather the evidence.
- Name the constraint.
- Protect the boundary.
- Record unknowns.



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Energy path and decision boundary



Every arrow is a boundary. Boundaries need ratings, protection, isolation, and documentation.

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Decision checks

Decision	Evidence	Reject if
Keep a design choice	Measured or observed site/load data	It only sounds good
Reject a shortcut	Safety or equipment boundary	It hides a hazard
Carry unknown forward	Named verification step	It becomes fake certainty
Approve handoff	Next person can act safely	It depends on mind reading

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Guided example

Model one complete thinking path.

- Start with the client/job statement.
- Add one measured or observed fact.
- Make one design decision.
- Name one safety boundary.
- Name one unknown and how to verify it.



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Team roles

Three teams of two, roles rotate after the first pass.

- Designer: proposes the first path.
- Verifier: challenges evidence and limits.
- Recorder/handoff lead: writes what the next person needs.
- In a pair, one person may hold two roles, then swap.



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Lab mission

String Design Paper Lab

- Teams compare three paper string options against controller limits and shade constraints.



This is the work block. Keep it practical and evidence-based.

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Scenario options

Assign one, or let teams choose if the room can handle democracy today.

- Four 22 Vmp / 27 Voc training modules feeding a 100 V controller.
- Two shaded morning modules and two unshaded modules on a shed roof.
- Long ground-rack wire run where voltage drop and current both matter.



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Lab procedure

Run the same loop every day.

- Read the module and controller cards.
- Build three possible string layouts using cards.
- Calculate approximate operating voltage/current and open-circuit voltage.
- Mark shade exposure from Day 2 scenario notes.
- Reject any option that violates a limit or ignores shade evidence.



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Acceptance criteria

A complete answer has five parts.

- Match the need.
- Show the path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.



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Safety boundary

No training objective is improved by pretending energy is harmless.

- Use only instructor-approved equipment and procedures.
- Treat PV and batteries as energy sources.
- Stop when measurements do not make sense.
- Follow manufacturer instructions and program safety procedures.

Stop, isolate, verify, ask. The glamorous life.



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Instructor probe questions

Use these when teams get vague.

- What evidence supports that?
- What would make this design fail?
- Which boundary protects the person or equipment?
- What unknown are you carrying forward?
- What would the next crew need to know?



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Peer verification

Peers check the handoff, not the personality of the speaker. Humanity struggles here.

- Can I identify the job?
- Can I follow the path?
- Can I name the constraint?
- Can I see the safety boundary?
- Can I find the unknowns?



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Common failure modes

Correct these early.

- Component list with no job.
- Guess presented as fact.
- Missing safety boundary.
- No units or runtime.
- Handoff too vague to use.



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Evidence language

Require technical sentences.

- Observed: what you saw.
- Measured: value with unit or clear method.
- Assumed: guess being carried forward.
- Verified: checked against source, procedure, or measurement.



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Handoff structure

Make the next person safer.

- Scope: what job is being solved.
- Decision: what was chosen or rejected.
- Evidence: why.
- Boundary: what must be protected.
- Unknowns: what still needs verification.



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Exit ticket

Individual close-out

- What made a string option unacceptable today: equipment limit, shade reality, or missing information?



Collect before reset so the day leaves a paper trail. Weirdly useful.

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Day-to-day continuity

This is one system, not seven disconnected worksheets.

- PV102 Day 5 turns daily energy need into storage, autonomy, and battery-protection decisions.
- Carry forward site notes, load tables, diagrams, and unknowns.
- Do not erase uncertainty. Resolve it.



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Final reminder

What professionals do

- Define the job.
- Verify the path.
- Protect the boundary.
- Document the truth.



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Close

- String design follows limits, temperature, and shade reality.



Reset tools. Collect packets. Archive the handoffs.

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